

Substituent Effects in Parallel-Displaced π – π Stacking Interactions: Distance Matters

Leslie-Joana Riwar, Nils Trapp, Bernd Kuhn, and François Diederich*

Dedicated to Professor J. Fraser Stoddart on the occasion of his 75th birthday

Abstract: Host–guest systems with Rebek imide type receptors and a 2,6-di(isobutyramido)pyridine ligand were employed to investigate substituent effects in parallel-displaced π – π stacking interactions. Changing the intermolecular distance between the *para* substituent on the aromatic platform of the receptor and the pyridine ring of the guest results in a strongly different substituent effect. With a short ethyne-1,2-diyl spacer between the Rebek imide and the aromatic platform, partial overlap of substituent and guest stabilizes the π – π stacking interactions independent of the electronic nature of the substituent (Wheeler–Houk model). When the substituent is shifted further away by using a buta-1,3-diyne-1,4-diyl spacer, direct, through-space interactions between substituent and guest are prevented. A linear correlation between $\log K_a$ (K_a = association constant) and the Hammett substituent constant σ_{para} is observed, confirming predictions by the Hunter–Sanders model experimentally.

Noncovalent interactions between aromatic rings play an eminent role in biological and chemical recognition.^[1,2] Insight into these interactions has been growing immensely over the past decades. Yet the mechanism by which substituents on aromatic rings affect parallel π – π stacking interactions remains a much debated topic of experimentally and theoretically oriented publications.^[3–8] Two characteristic models have emerged over time, explaining substituent effects in substituted benzene dimers. In the Hunter–Sanders model, electron-withdrawing substituents decrease the π -electron density in the substituted benzene and thereby improve the interaction. Electron-donating substituents, on the other hand, push electron density into the substituted benzene, thereby increasing the electrostatic repulsion between the stacked benzene derivatives.^[6] In the Wheeler–Houk model, the electrostatic influence of the substituent on the π -cloud is overshadowed by direct, through-space interactions based on the local dipoles of the substituent.^[7] Both

effects were evident in a theoretical study by Arnstein and Sherrill.^[8d] They investigated substituent effects on substituted, parallel-displaced benzene dimers as a function of the substituent displacement. Depending on the position of the substituent, either the effect of the π -cloud on the electronic structure or direct interactions with the substituent were prevalent in determining the stacking strength.

Rebek imide derivatives have successfully been employed to quantify weak intermolecular interactions.^[9] Recently, we reported a geometrically well-defined host–guest system with Rebek imide receptors **1** and di(isobutyramido)pyridine **2** as the substrate to investigate substituent effects on π – π stacking interactions (Figure 1, left).^[10] Guest binding occurs through

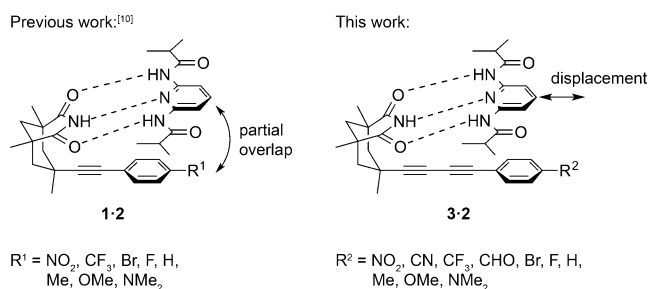


Figure 1. Schematic representation of Rebek imide type model systems. Left: The recently reported receptor–guest complex **1-2** with partial overlap of the guest and the benzene platform substituent in the *para* position.^[10] Right: The new receptor–guest complex **3-2** with an elongated linker to shift substituents away from the guest.

a triple hydrogen-bonding array to the imide scaffold of the receptor, thereby enabling parallel stacking between the aromatic platform of **1** and the pyridine core in **2**. X-ray co-crystal structures revealed an ideal centroid-to-centroid distance of 3.5 Å (Figure 2). They also showed a partial overlap between the *para* substituents on the aromatic platform of receptor **1** and the pyridine ring of **2** (3.3 Å). At this close proximity, various substituents had a stabilizing effect on π – π stacking independent of their electronic nature, which is in agreement with the Wheeler–Houk model.^[7] Hence, no linear correlation between the complexation strength and the Hammett substituent constant σ_{para} was observed (Figure 3, top).^[11]

Herein, we present the new host–guest system **3-2** with the linker between the aromatic platform and the Rebek imide elongated from ethyne-1,2-diyl (in **1**) to buta-1,3-diyne-1,4-diyl (in **3**; Figure 1, right). We highlight the key differences between the experimentally measured substituent effects in

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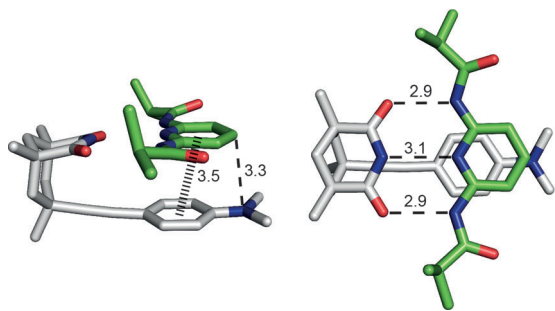
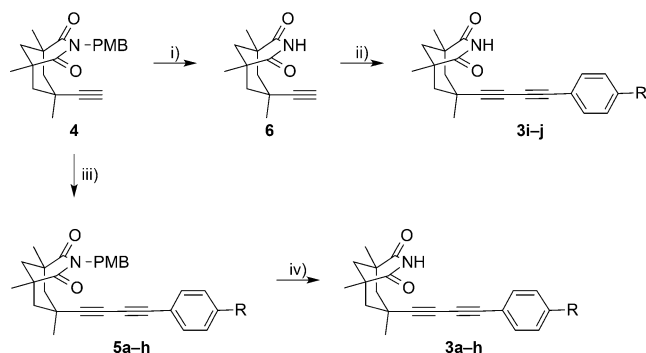


Figure 2. X-ray co-crystal structure of complex **1·2** ($T = 100$ K, space group: monoclinic $P2_1/c$), side view (left) and top view (right).^[10] Gray skeleton: receptor **1** with $R = \text{NMe}_2$, green skeleton: guest **2**. N blue, O red. Distances are given in Å. Hydrogen atoms are omitted for clarity.

the two systems, which result from changing the distance between the *para* substituent and the pyridine ring.

The preparation of the modified receptors **3a–j** is presented in Scheme 1. PMB-protected imide **4** was synthesized according to a literature procedure.^[9a,e,10,12] Cadiot–

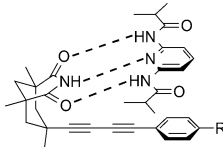


Scheme 1. Synthesis of target receptors **3a–j**. Reagents and conditions: i) CAN, MeCN/H₂O (10:1), 0 to 22 °C, 83%; ii) (*p*-iodoethynyl)-phenyl- R ($R = \text{OMe}, \text{NMe}_2$), [Pd(PPh₃)₂Cl₂], CuI, NEt₃, 22 °C, 11–36%; iii) (*p*-iodoethynyl)phenyl- R ($R = \text{NO}_2, \text{CN}, \text{CF}_3, \text{CHO}, \text{Br}, \text{F}, \text{H}, \text{Me}$), [Pd(PPh₃)₂Cl₂], CuI, NEt₃, 22 °C, 11–78%; iv) CAN, MeCN/H₂O (10:1), 0 to 22 °C, 52–93%. CAN = ceric ammonium nitrate, PMB = *para*-methoxybenzyl.

Chodkiewicz cross-coupling with the respective iodoacetylenes afforded the PMB-protected receptors **5a–h**. Target compounds **3a–h** were obtained after CAN-mediated deprotection. For receptors **3i** and **3j**, PMB deprotection had to precede the cross-coupling owing to decomposition in the final step (see the Supporting Information for details).

Complexation of di(isobutyramido)pyridine **2** by the new receptors **3a–j** was investigated under identical conditions to the previous study with **1**. ¹H NMR titration studies were performed in a (CDCl₃)₂ solution at 298 K at a constant host concentration (5 mM) and increasing guest concentration (up to 100 mM). Association constants K_a were determined by nonlinear curve fitting of the large complexation-induced downfield shift of the imide N–H resonance at fast host–guest exchange (Table 1).^[13] Similar results were obtained when following the upfield shifts of the *ortho* and *meta* protons on

Table 1: K_a [M^{−1}], log K_a values, and Gibbs binding free energies ΔG [kJ mol^{−1}] for 1:1 complexes of **3·2** ((CDCl₃)₂, 298 K, [host] = 5.0 mM).



R	Complex	K_a [M ^{−1}] ^[a]	log K_a	ΔG [kJ mol ^{−1}] ^[b]
NO ₂	3a·2	91.1	1.96	−11.18
CN	3b·2	90.7	1.96	−11.17
CF ₃	3c·2	87.6	1.94	−11.08
CHO	3d·2	85.8	1.93	−11.03
Br	3e·2	80.2	1.90	−10.86
F	3f·2	83.1 ± 4.6 ^[c]	1.92	−10.95
H	3g·2	72.4	1.86	−10.61
Me	3h·2	68.0	1.83	−10.45
OMe	3i·2	73.9	1.87	−10.66
NMe ₂	3j·2	64.2	1.81	−10.31

[a] Averaged K_a values from Table S1. [b] Derived from $\Delta G = -RT \ln K_a$.

[c] Standard deviation from four independent measurements.

the platform of receptor **3** or the ¹⁹F shift in complex **3f·2**, albeit with reduced accuracy (Table S4 and Figure S11 in the Supporting Information). The observed complexation-induced upfield shifts of the phenyl protons are small ($\Delta\delta_{\text{max},3·2,\text{meta}} = -0.5$ ppm, $\Delta\delta_{\text{max},3·2,\text{ortho}} = -0.2$ ppm; *ortho*, *meta* to the substituent R) compared to the downfield shift of the N–H resonance ($\Delta\delta_{\text{max,NH}} = +4.1$ ppm) and are therefore more prone to error. Interestingly, the observed upfield shifts of the aromatic protons correspond to the observed shifts in system **1·2**, but in an opposing manner. In that system, the larger upfield shift was observed for the *ortho* proton ($\Delta\delta_{\text{max},1·2,\text{meta}} = -0.2$ ppm, $\Delta\delta_{\text{max},1·2,\text{ortho}} = -0.5$ ppm), indicating a stronger overlap of the aromatic platform with the pyridine ring of guest **2**. The ring current effect of the pyridine clearly influences the upfield shift of the subjacent aromatic protons at different displacement.

The overall magnitudes of the Gibbs binding free energies ΔG of both systems **1·2** and **3·2** are comparable and rather small owing to conformational entropy losses upon complexation, as discussed in the previous publication.^[10] Remarkably, however, the larger displacement of the substituent in **3** results in a good linear correlation between log K_a and σ_{para} (Figure 3, bottom, $r^2 = 0.89$). This correlation indicates an electrostatic interaction between the substituted phenyl ring and the pyridine guest, as predicted by the Hunter–Sanders model. Electron-withdrawing substituents enhance the π – π stacking relative to the unsubstituted assembly (**3g·2**), whereas electron-donating substituents cause a destabilization.^[14] This finding is in stark contrast to the results obtained for the previously studied system **1·2**.^[10] For the latter, no such correlation was observed (Figure 3, top), and various substituents strengthened the π – π stacking, independent of their electronic nature. In that system, partial overlap of substituent and guest allowed direct, through-space interactions between the substituent and the pyridine ring, as predicted by the Wheeler–Houk model. Our experimental results support the theoretical work of Arnstein and Sherrill, where substituents have complex effects on the interaction energy if

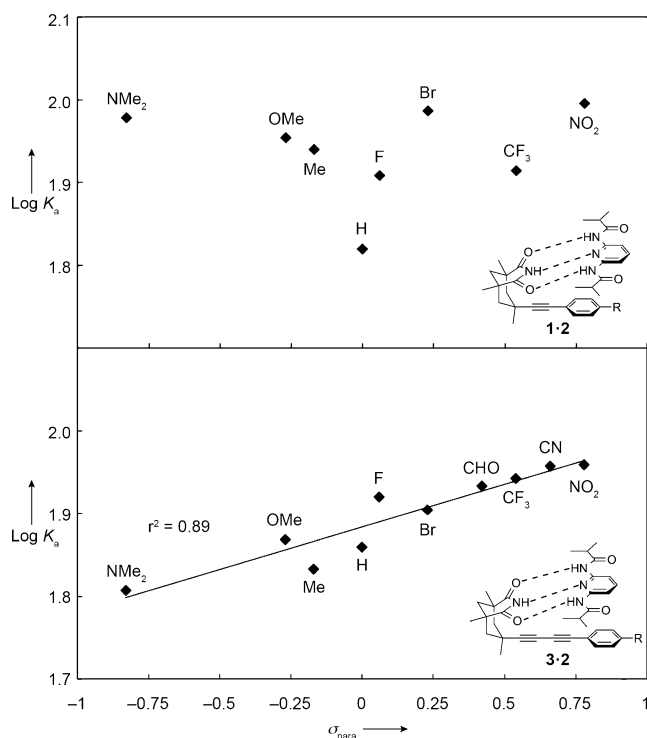


Figure 3. Correlation plot of $\log K_a$ and σ_{para} in (CDCl₂)₂ at 298 K. Top: Values from the previously published host-guest system **1-2**.^[10] Bottom: New host-guest system **3-2**.

they overlap with the other ring.^[8d] For a large displacement (see **3-2**), the interaction energy is mainly influenced by the electronic structure of the aromatic ring imparted by the substituent, not by the substituent itself.

For the earlier system **1-2**, all structural investigations in solution and in the solid state supported a π - π stacking alignment of the pyridine guest and the aromatic ring in the Rebek imide platform.^[10] As both π - π stacking and edge-to-face interactions between these rings would yield a similar linear correlation with the Hammett substituent parameter, extensive structural analysis in the crystalline state and in solution, accompanied by a computational study, was conducted for the extended system **3-2**, confirming the prevalence of the π - π stacking geometry of the complexes in solution.

Single crystals of differently substituted receptors **3** and complexes **3-2** were obtained for X-ray diffraction, and the solid-state structures are discussed in the Supporting Information (Section S4).^[15,16] Both host-guest geometries, namely π - π stacking and edge-to-face alignment of the pyridine and platform arene rings, were observed for the co-crystal structures, in contrast to the findings for **1-2**. The greater flexibility of the buta-1,3-diyne-1,4-diyl bridge enables a larger diversity of packing modes in the crystal with different stoichiometries and host-guest orientations, which are also driven by strong hydrogen-bonding interactions, pairing hydrogen-bond donors and acceptors.

The co-crystal structure of **3c-2** shows that host-guest association occurs through a triple hydrogen-bonding array between the imide moiety of the host and the guest ($d(O\cdots N) = 3.0$ Å, $d(N\cdots N) = 3.1$ Å; Figure 4b, top right).

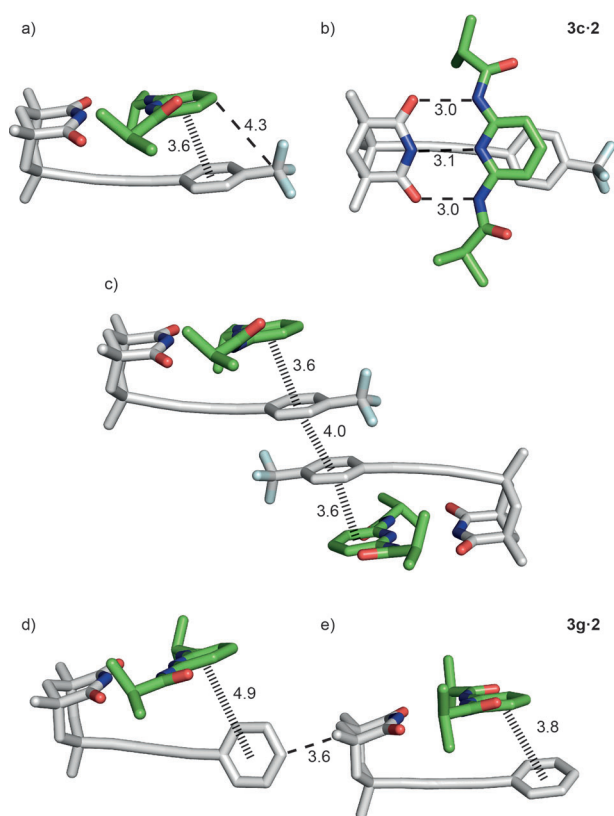


Figure 4. X-ray co-crystal structure of **3c-2**, side view (a), top view (b), and crystal packing (c). Co-crystal structure of **3g-2** showing two host-guest complex conformations, namely edge-to-face (d) and parallel-stacked (e). Gray skeleton: receptor, green skeleton: guest. F cyan, N blue, O red. Distances are given in Å. Hydrogen atoms are omitted for clarity.^[16]

Upon complexation, the acetylene linker bends towards the imide ($C-C\equiv C$ angles from 174° to 178°) to bring the aromatic platform closer to the guest. This allows for an ideal parallel-displaced stacking interaction for system **3c-2** with a centroid-to-centroid distance of 3.6 Å (Figure 4a). Compared to the previous system **1-2** (Figure 2), an increased distance between the *para* substituent and the outer vertex of **2** (4.3 Å) is observed, which prevents any local, direct interactions with the substituent. The crystal packing reveals a parallel aromatic interaction network involving two complexes **3c-2**, where the dipoles of the receptor platforms are aligned in a favorable antiparallel fashion (Figure 4c). In the ensemble of co-crystal structures (Section S4), the platform arene rings of neighboring complexes show both π - π stacking and edge-to-face alignments.

The co-crystal structure of **3g-2** (Figure 4d,e and Section S4.4.3) contains two conformations of the host-guest complexes, the first one with intermolecular edge-to-face and the second one with parallel-displaced aromatic interactions between host and guest ($d(\text{centroid-to-centroid}) = 4.9$ and 3.8 Å, respectively). In the edge-to-face conformation, guest **2** is pushed away to allow for an increased distance between the guest and the platform of the receptor. The crystal packing of several co-crystal structures reveals that rotation of certain phenyl rings to the edge-to-face conformation enables

C–H $\cdots\pi$ interactions and further arene–arene interactions to neighboring receptors (see Section S4.4).^[17]

The binding mode of system **3-2** in solution was first investigated by ¹H steady-state NOE experiments (Figure 5

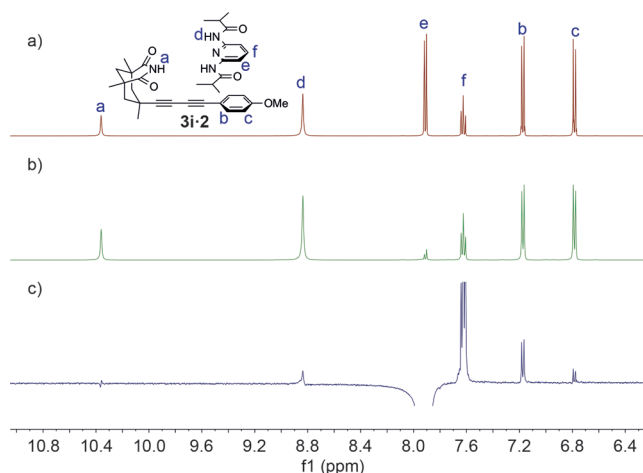


Figure 5. 500 MHz ¹H NOE spectra of **3i-2** in (CDCl₂)₂ at 298 K; [H] = 38.2 mM, [G] = 43.0 mM. A) Reference spectrum, B) irradiated spectrum, C) difference spectrum.

and Section S3). Irradiation on the frequency of the aromatic protons e in position 3 of the pyridine core shows a clear NOE for the aromatic protons b and c, which are in *meta* and *ortho* position to the substituent, respectively. Hence, it can be assumed that pyridine guest **2** and the aromatic platform of receptor **3** are in close proximity owing to arene–arene interactions. The 1:1 stoichiometry was additionally confirmed by Job plot analysis (Figure S21).^[18]

In the solution analysis, however, no evidence for edge-to-face interactions in our system was found. The π – π stacking geometries of several systems in solution have been evidenced by using ring-current-induced changes in chemical shifts.^[3a–c,j,6c,19–21] In π – π stacking complexes, the protons at the periphery of the stack usually encounter upfield shifts on the order of –0.2 to –0.5 ppm, as we observed for both systems **1-2** and **3-2** (see above). No such upfield shifts are observed for the face ring in an edge-to-face assembly. In a reverse-titration study with di(isobutyramido)pyridine **2** at constant concentration (4 mM), we measured a substantial complexation-induced upfield shift that is due to the intermolecular shielding of the aromatic pyridine protons (Section S2.1.4). For example, the proton in position 4 moves upfield during the reverse titration by $\Delta\delta_{\text{max,obs}} = -0.32$ ppm ($\Delta\delta_{\text{sat,calc}} = -0.35$ ppm; Figure S19). Hence, regardless of the titration mode, all involved aromatic protons show an upfield shift upon complexation, indicating a parallel stacked geometry in solution.^[3b,c,j,6c,20] Indeed, if our system formed an intermolecular edge-to-face interaction, as suggested by some of the X-ray co-crystal structures (Section S4.4), a downfield change in chemical shift for the pyridine protons would be expected.^[3a,21] The parallel stacking geometry was further supported by quantum-mechanical geometry optimizations (Section S5), where all optimizations for system **3i-2** converged

to an energetically more favorable parallel-displaced orientation. Therefore, we can conclude with confidence that parallel-displaced π – π stacking is the dominant interaction in solution.

In summary, we have reported the new Rebek imide type receptor model system **3-2**, which, together with the previously described host–guest system **1-2**,^[10] for the first time enabled an experimental determination of the distance dependence of substituent effects on parallel π – π stacking interactions. In the host–guest system **1-2**, the short ethyne-1,2-diyl spacer generates partial overlap between the phenyl platform substituent in the *para* position and the guest di(isobutyramido)pyridine **2**. As predicted by the Wheeler–Houk model, the introduction of any substituent resulted in stabilization by direct, through-space interactions. In the new system **3-2**, the elongated buta-1,3-diyne-1,4-diyl spacer shifts the substituent away from the bound pyridine guest, which prevents direct interactions between substituent and guest. A linear correlation between $\log K_a$ and the Hammett substituent constant σ_{para} was observed, with electron-withdrawing substituents leading to a stabilization of the complex and electron-donating substituents destabilizing the interaction. This finding indicates an electrostatic influence of the substituent on the aromatic platform of the receptor, as predicted by the Hunter–Sanders model. The distance between the substituent and the intermolecularly interacting aromatic ring clearly matters when analyzing substituent effects on aromatic π – π stacking interactions.

Experimental Section

General procedures for the Cadiot–Chodkiewicz cross-coupling and PMB deprotection are described here. Further synthetic details and characterization can be found in the Supporting Information.

General procedure 1 (GP1): Cadiot–Chodkiewicz cross-coupling: Terminal alkyne **4** or **6** (1.0 equiv), the corresponding (iodoethynyl)benzene derivative (1.5 equiv), CuI (0.1 equiv), and [Pd(PPh₃)₂Cl₂] (0.05 equiv) under an Ar atmosphere were suspended in freshly distilled NEt₃ and stirred for 16 h. The mixture was diluted with Et₂O and filtered over Celite. After evaporation of the filtrate in vacuo, the residue was purified by column chromatography (SiO₂).

General procedure 2 (GP2): CAN-mediated PMB deprotection: PMB-protected imide **4** or **5** (1.0 equiv) was dissolved in MeCN/H₂O (10:1, 10 mm) at 0°C. The solution was treated with CAN (10.0 equiv), stirred for 15 min at 0°C, warmed to 22°C, and stirred until completion (monitored by TLC). The mixture was carefully treated with sat. aq. NaHCO₃ solution, and the organic solvent was removed in vacuo. The residual aqueous layer was extracted with CH₂Cl₂ (3 $\times$). The combined organic layers were dried over Na₂SO₄ and evaporated in vacuo. The residue was purified by column chromatography (SiO₂).

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Conflict of interest

The authors declare no conflict of interest.

Keywords: host–guest systems · Rebek imides · substituent effects · supramolecular chemistry · π – π stacking interactions

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- [15] Co-crystals were obtained from a solution containing an excess of **2** (up to tenfold). Only a small share of the crystallized samples contained co-crystals owing to predominant dimerization of guest **2**.
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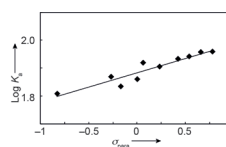
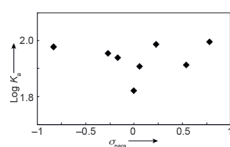
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Communications

VIP

Host–Guest Systems

L.-J. Riwar, N. Trapp, B. Kuhn,
F. Diederich*     Substituent Effects in Parallel-Displaced
 π – π Stacking Interactions: Distance
Matters

Distance matters: Substituents have a different effect on π – π stacking interactions depending on their position. Stacking interactions in Rebek imide model systems with partial substituent

overlap follow the Wheeler–Houk model whereas a larger displacement supports the Hunter–Sanders model. Two controversial theoretical predictions were confirmed experimentally.